Customisable Routing Engine

Saskia Jessop (2007261) CS344 Presentation

Control

What is the optimisation parameter?

How are different priorities weighted?

What is the target demographic?

komoot Discover Route planner Features Shop English Sign up or log in

Route Planner

Save Tour

Search For Place Or Address

Sport Running

Fitness In Good Shape

Route Type One Way

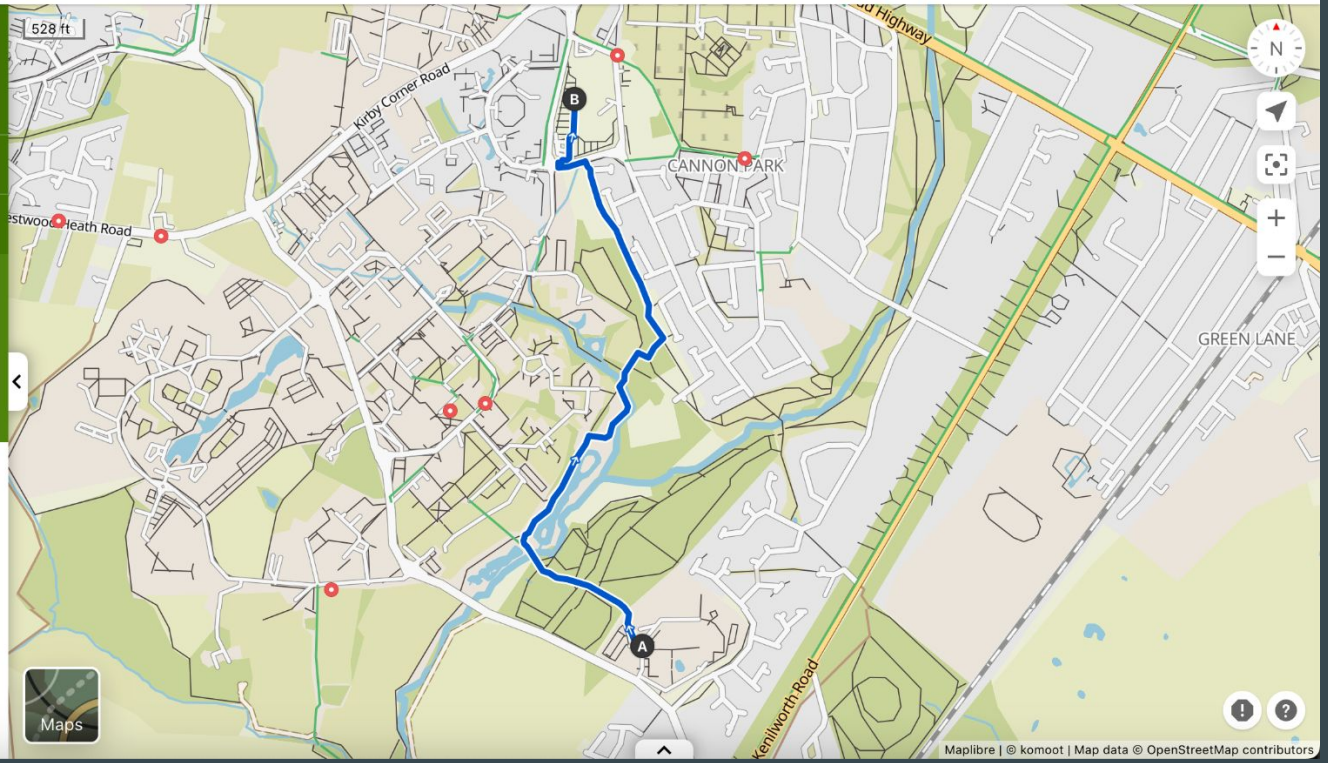
A Waypoint A

B Waypoint B

00:13 | 1.22 mi | 25 ft | 50 ft

Easy Easy run. Great for any fitness level. Mostly paved surfaces. Suitable for all skill levels.

WAYTYPES



OpenStreetMap | Edit | History | Export | GPS Traces | User Diaries | Communities | Copyright | Help | About | Log In | Sign Up

52.3748, -1.5531

52.38732, -1.55556

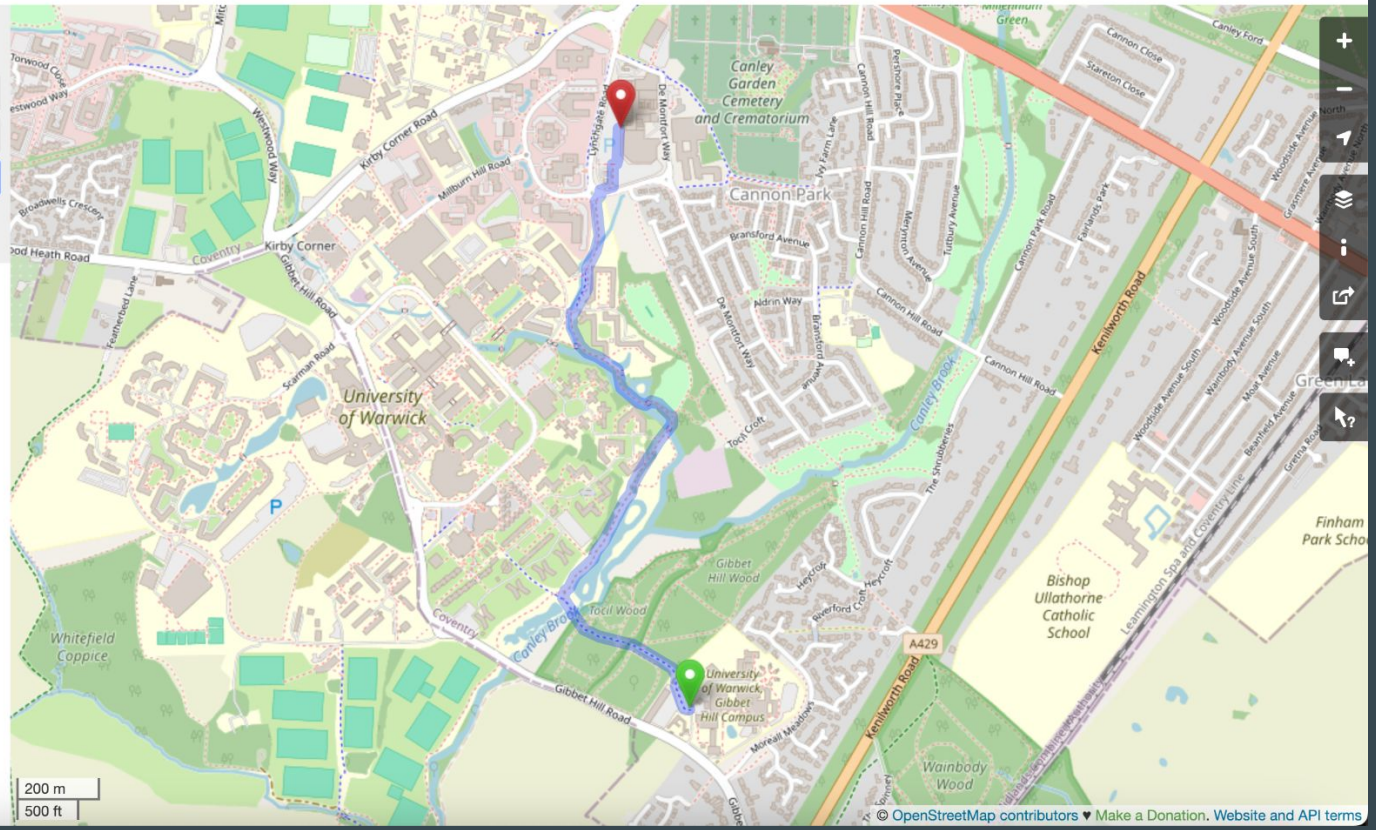
Foot (OSRM) | Go

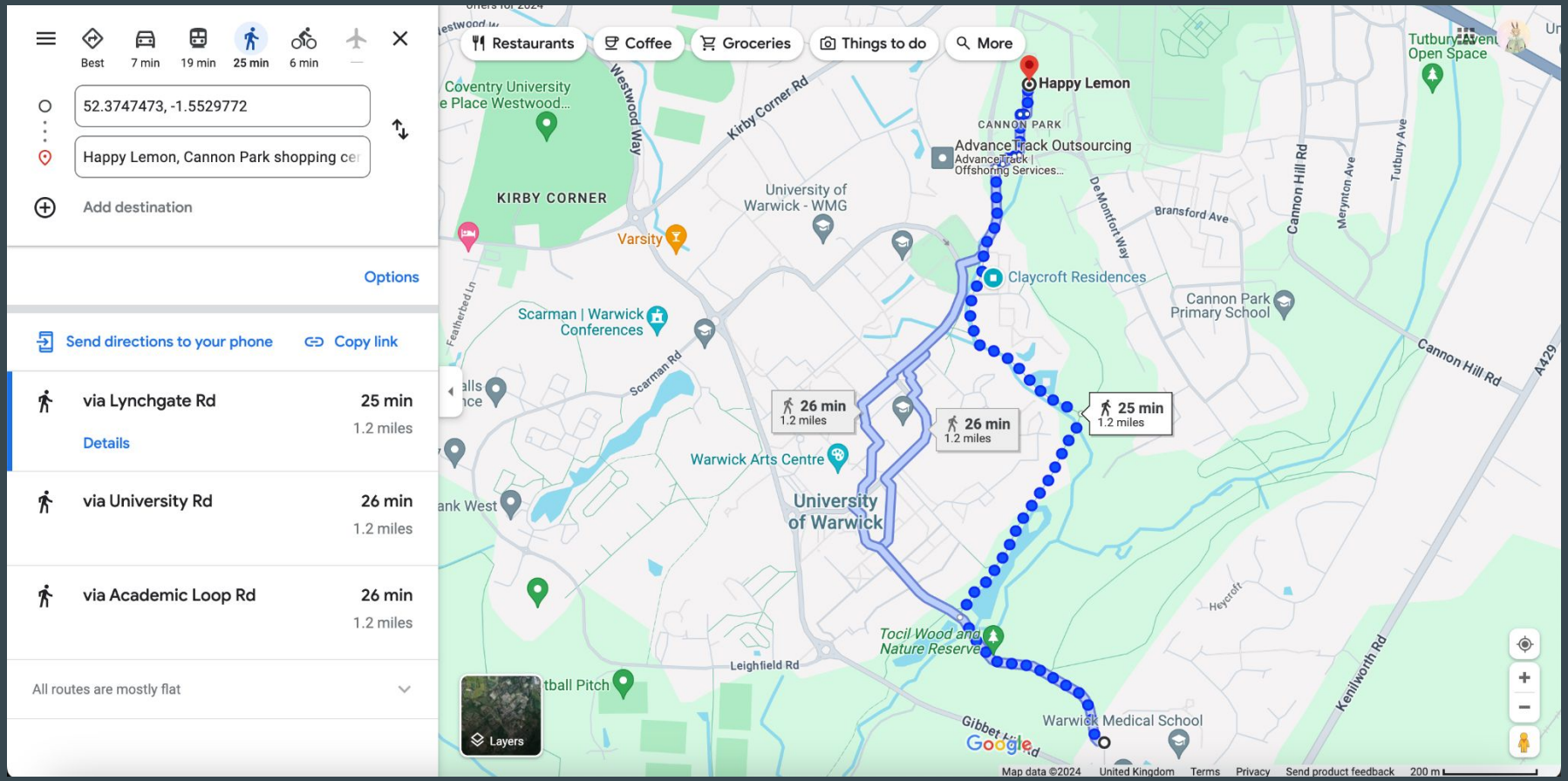
Reverse Directions

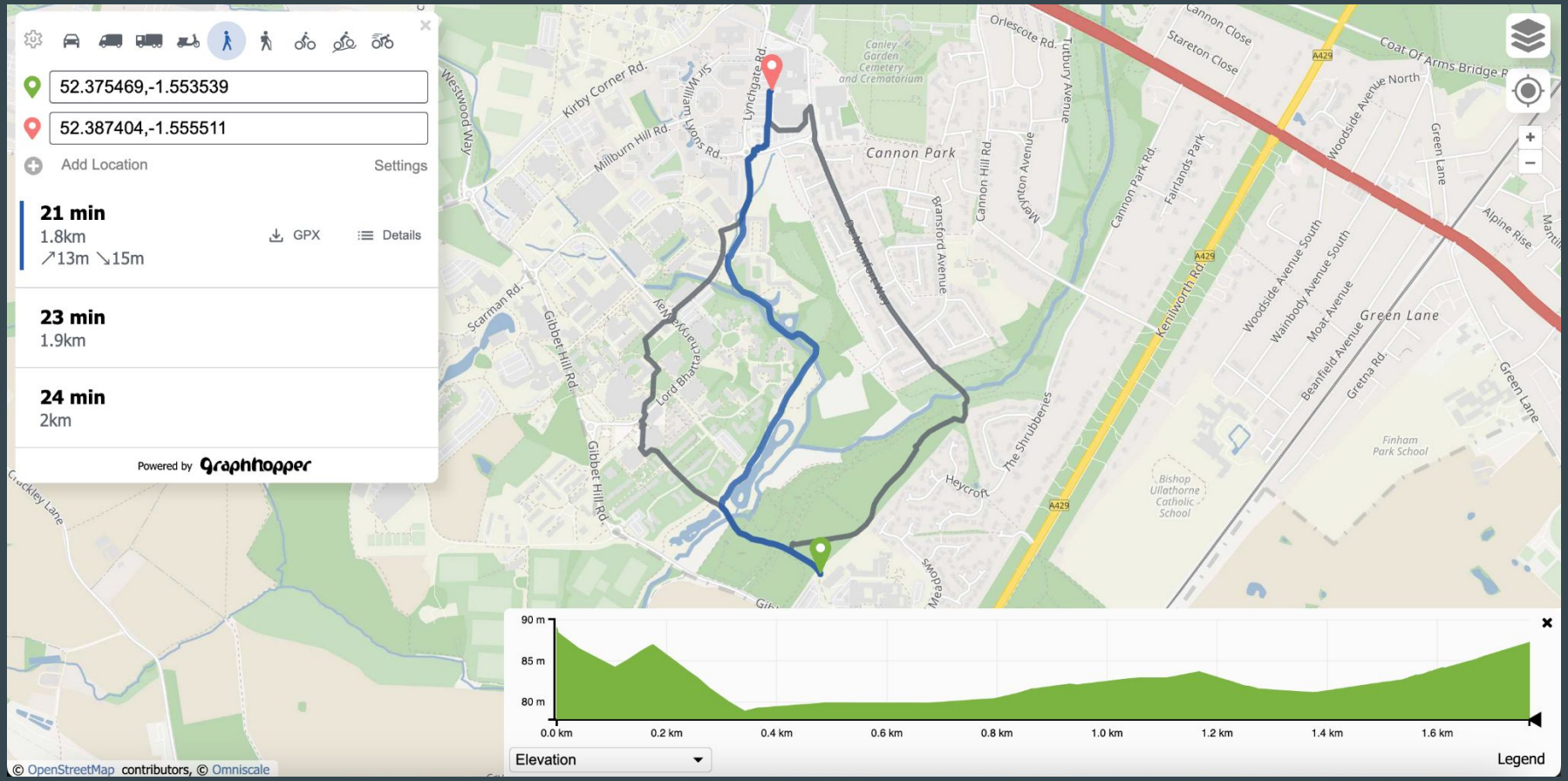
Directions

Distance: 1.9km. Time: 0:25.

1. Start on unnamed road	400m
2. Turn right onto unnamed road	500m
3. Turn left onto unnamed road	400m
4. At the end of the road turn right onto unnamed road	10m
5. Turn left onto unnamed road	10m
6. At the fork turn right onto unnamed road	100m
7. Slight right onto unnamed road	180m
8. Turn left onto unnamed road	10m
9. At the end of the road turn right onto unnamed road	40m
10. Turn left onto unnamed road	50m




Best
7 min
19 min
25 min
6 min
52.3747473, -1.5529772
Happy Lemon, Cannon Park shopping cer
Add destination
Options
Send directions to your phone
Copy link
via Lynchgate Rd
25 min
1.2 miles
Details
via University Rd
26 min
1.2 miles
via Academic Loop Rd
26 min
1.2 miles
All routes are mostly flat
Restaurants
Coffee
Groceries
Things to do
More
Happy Lemon
CANNON PARK
Advance Track Outsourcing
Advance Track | Offshoring Services...
Coventry University
Place Westwood...
KIRBY CORNER
Westwood Way
Kirby Corner Rd
University of Warwick - WMG
Varsity
Claycroft Residences
Scarman | Warwick Conferences
Featherbed Ln
Scarman Rd
De Montfort Way
Bransford Ave
Cannon Hill Rd
Merynton Ave
Tutbury Ave
Tutbury Avenue Open Space
Cannon Park Primary School
Cannon Hill Rd
A429
26 min
1.2 miles
26 min
1.2 miles
25 min
1.2 miles
Warwick Arts Centre
University of Warwick
Heycroft
Tocil Wood and Nature Reserve
Leighfield Rd
Kenilworth Rd
Warwick Medical School
Gibbet Hill Rd
Layers
Map data ©2024
United Kingdom
Terms
Privacy
Send product feedback
200 m


52.375469,-1.553539
52.387404,-1.555511
Add Location
Settings
21 min
1.8km
↗13m ↘15m
GPX
Details
23 min
1.9km
24 min
2km
Powered by graphhopper
Cannon Park
Canley Garden Cemetery and Crematorium
Finham Park School
Bishop Ullathorne Catholic School
Green Lane
Elevation
Legend
90 m
85 m
80 m
0.0 km
0.2 km
0.4 km
0.6 km
0.8 km
1.0 km
1.2 km
1.4 km
1.6 km
© OpenStreetMap contributors, © Omniscale

Control

Discrepancies in different engines due to

- map data
- underlying algorithm

which are inaccessible to general audiences

There exist few context dependent engines

Existing Systems

Tailored to specific demographics:

HOBBIT - Avoids other hikers through Wi-Fi signals

Maaret Posti, Johannes Schöning, and Jonna Häkkilä. 2014. Unexpected journeys with the HOBBIT: the design and evaluation of an asocial hiking app. In Proceedings of the 2014 conference on Designing interactive systems (DIS '14). Association for Computing Machinery, New York, NY, USA, 637–646. https://doi.org/10.1145/2598510.2598592

Existing Systems

Tailored to specific demographics:

HOBBIT - Avoids other hikers through Wi-Fi signals

Autobahn - Finds scenic routes from Flickr image data

Nina Runge, Pavel Samsonov, Donald Degraen, and Johannes Schöning. 2016. No more Autobahn! Scenic Route Generation Using Googles Street View. In Proceedings of the 21st International Conference on Intelligent User Interfaces (IUI '16). Association for Computing Machinery, New York, NY, USA, 147–151. https://doi.org/10.1145/2856767.2856804

Existing Systems

Tailored to specific demographics:

HOBBIT - Avoids other hikers through Wi-Fi signals

Autobahn - Finds scenic routes from Flickr image data

Research on customisation:

"Trip Router with Individualized Preferences (TRIP): Incorporating Personalization into Route Planning"

Julia Letchner, John Krumm, and Eric Horvitz. 2006. Trip router with individualized preferences (TRIP): incorporating personalization into route planning. In Proceedings of the 18th conference on Innovative applications of artificial intelligence - Volume 2 (IAAI'06). AAAI Press, 1795–1800.

Problem Statement

- ➔ Make routing more accessible
- ➔ Allow ideal route conditions to be specified
  - Users can choose paths based on:
    - elevation,
    - surface type, and
    - wheelchair accessibility

Why GraphHopper?

Accessible API

Open-Source provider of structural metadata [1]

[1] Areeg Mostafa and Mohamed Mokbel. 2023. Towards Open-Source Maps Metadata. In Proceedings of the 31st ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '23). Association for Computing Machinery, New York, NY, USA, Article 31, 1–4. https://doi.org/10.1145/3589132.3625576

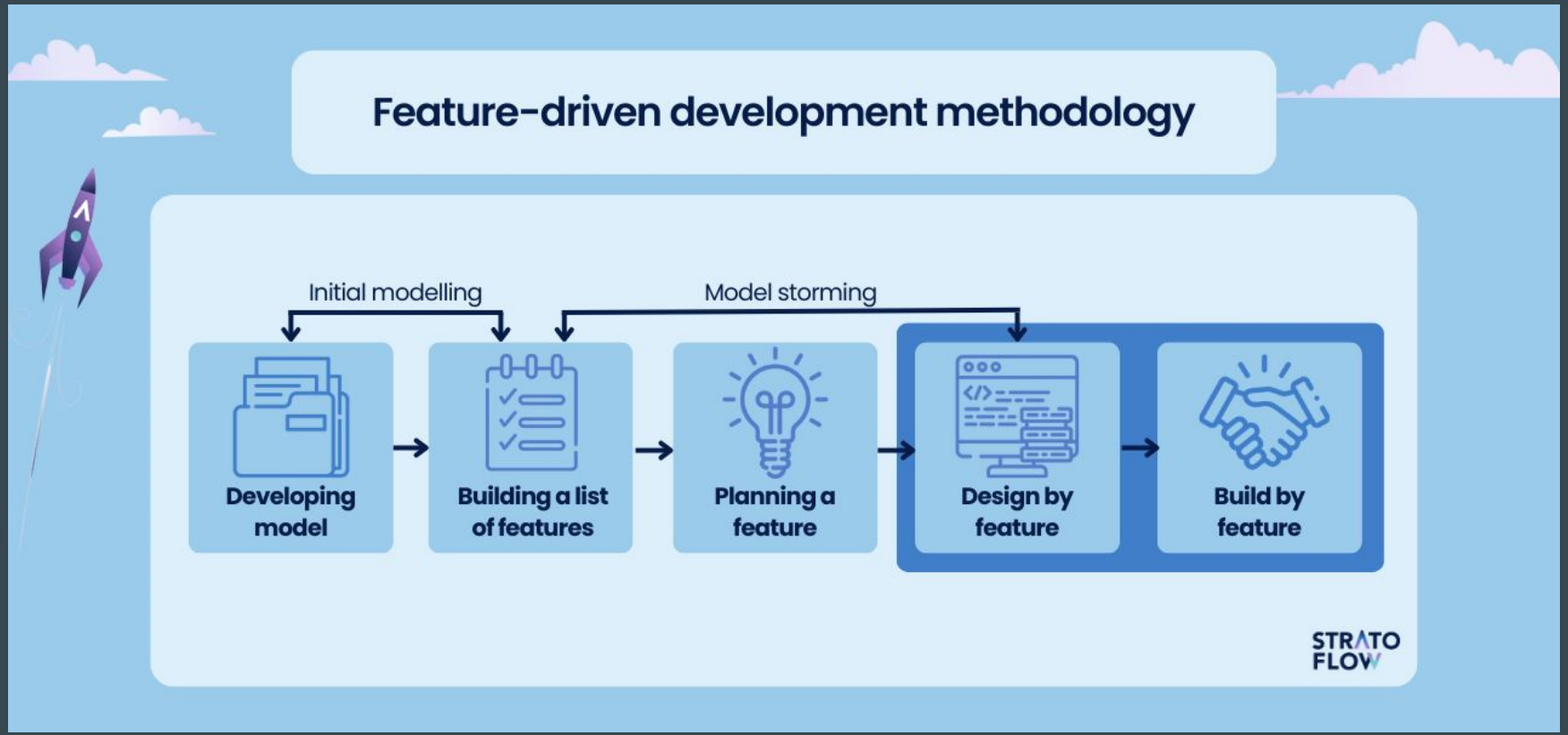
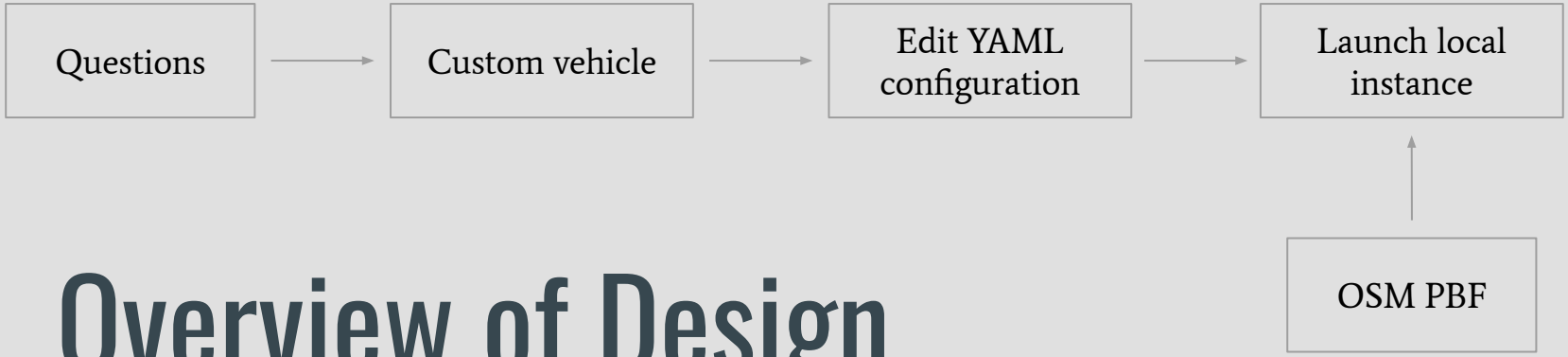


image source: https://stratoflow.com/software-development-methodologies/



Overview of Design

What does the final project look like?

$$edge_weight = edge_distance / (speed * priority) + edge_distance * distance_influence$$

constants for each vehicle

edge_weight = **edge_distance** / (**speed** * priority) + **edge_distance** * distance_influence

edge_weight = edge_distance / (speed * **priority**) + edge_distance * **distance_influence**

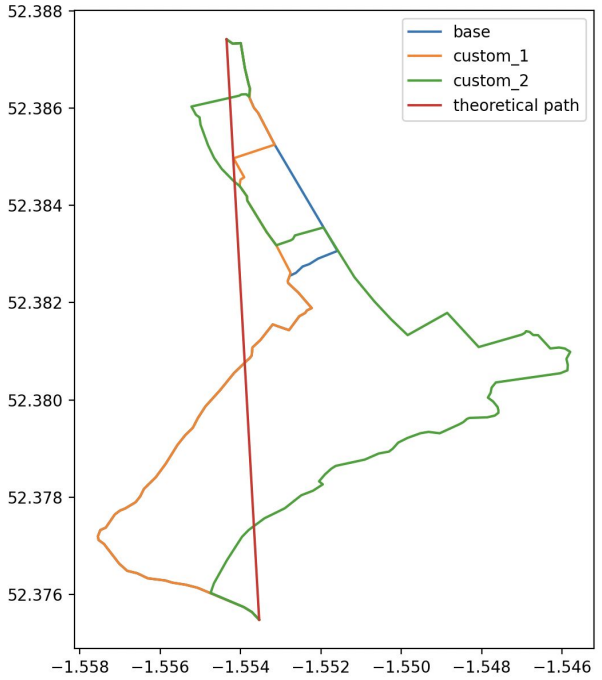
influenced by user's preferences

My approach: edit the priority value attached to those heuristic values

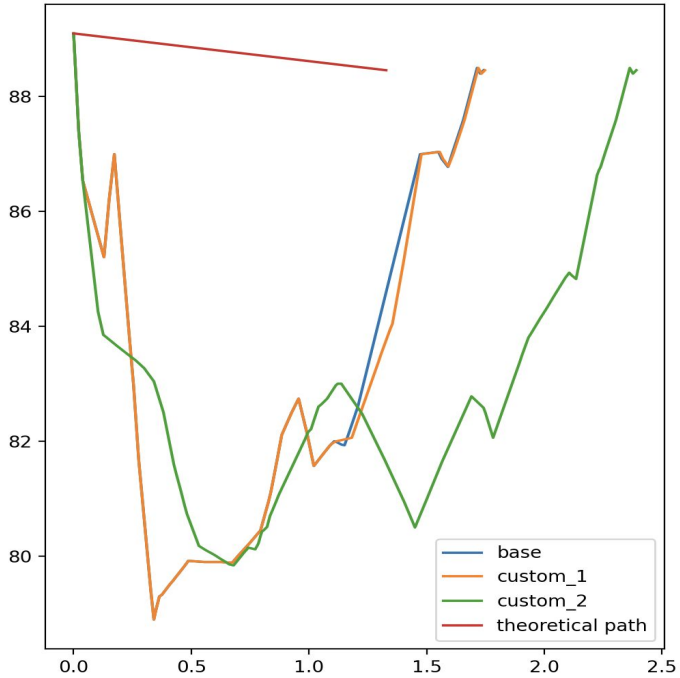
```
"custom_model": {
    "priority": [
        {
            "if": "max_slope > 1",
            "multiply_by": "0.3"
        }
    ]
}
```

```
"custom_model": {
    "priority": [
        {
            "if": "max_slope > 1",
            "multiply_by": "0.1"
        }
    ]
}
```

My approach: edit the priority value attached to those heuristic values



Plot of route (longitude against latitude)

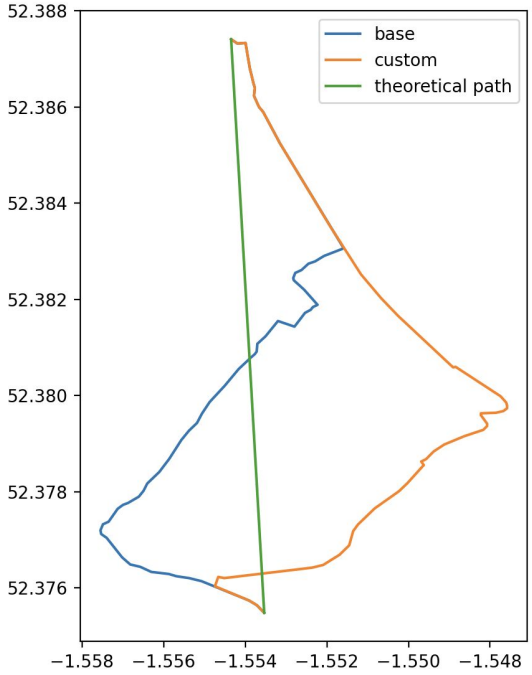


Plot of distance against elevation

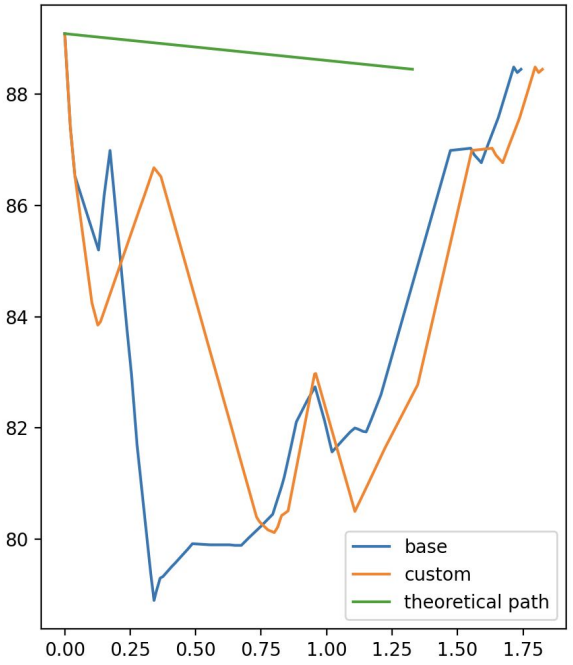
My approach: edit the priority value attached to those heuristic values

```
"custom_model": {
    "priority": [
        {
            "if": "surface != MISSING",
            "multiply_by": "0.4"
        },
        {
            "if": "max_slope > 6",
            "multiply_by": "0.5"
        }
    ]
}
```

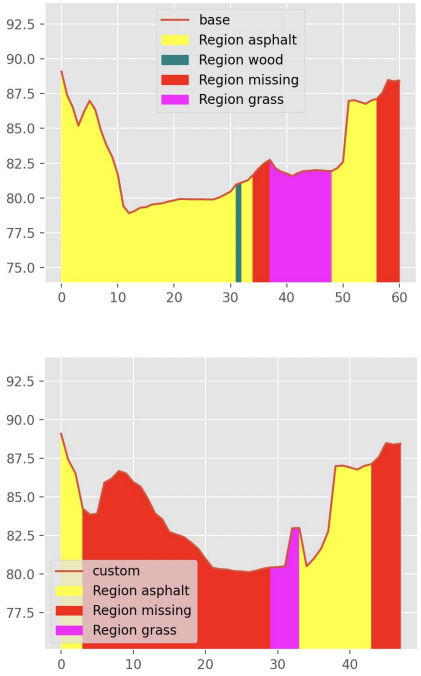
My approach: edit the priority value attached to those heuristic values



Plot of route (longitude against latitude)



Plot of distance against elevation



Plots of surface ranges

My approach: compare the efficiency ratios of different routes

Generate Base Requests

Heuristic Requests

- Slope (percentage, value)
- Surface (common, uncommon, value)
- Distance Influence (value)

Heuristic Analysis

- distance %↓ over slope range
- distance %↓ over surface range
- distance/time %↓

Generate Test Requests

Demo

Evaluation

- Most of the conditions for an MVP have been met
- Framework for testing has been established
- Website is launchable, so it is possible for users to reevaluate if route generated does not match their needs

- Had to re-evaluate schedule due to difficulties in automating the testing process
  - Wasn't able to access Route Optimization API on the standard package
- Need to test on more varied locations

Project Management

Menu | Project Specification | Review | Share | Submit | History | Layout | Chat

Code Editor | Visual Editor | Normal text | Recompile

- bibliography.bib
- final.tex
- Gantt.png
- inactive.png
- localhost.png
- progress.tex
- report.tex
- route.png
- spec.tex
- specification.tex
- term1.png
- term2.png
- tinysuccess.png
- readme.md

File outline

- Promises I made in Spec
- According to my Gantt Chart
- Problems I have encountered
- Promises I made in Report
- According to my Gantt Chart
- Problems I have encountered

```latex
\begin{enumerate}
    \item Evaluate results
    \begin{enumerate}
        \item Made 3 different files for analysing surface, slope, and distance\_influence. Will compare each factors deviation from a basic path to determine strength of opinion it matches with
        \item Finished distance\_influence evaluation code
        \begin{enumerate}
            \item checked against foot data and confirmed that distance\_influence has no effect on foot vehicle, therefore won't pose the question to foot
            \item still need to evaluate against car vehicle (TODO: download car data)
        \end{enumerate}
        \item Working on slope evaluation code
        \begin{enumerate}
            \item Realised that I need to work with max\_slope not average\_slope as it was allowing for dramatic downhills as well to fix the average
            \item Made a CSV file with all of the (slope change, distance change) pairs, found 5 invalid routes when the multiplication\_value was 0 for max\_slope > 3\% --> too strict of a weighting

            Could either warn users that 0 is too strict and offer alternative routes, or just don't allow it (TODO: needs further analysis to determine where the limit is)
            \item last 4 are always nan so implies that multiplication\_values = [0.6, 0.7, 0.8, 0.9] have little to no impact
            \item Coventry is quite flat so most of my elevation data was useless as the maximum slope tended to be in the 3-6\% range --> regenerating data with smaller percentage value inputs to match the data range of graphhopper which is 3\%
```

Week 6 (6th - 12th):

1. Solve 'path of shortest length' problem
   (a) Struggled a lot with setting up GraphHopper due to incomplete directions in the documentation. I noted down what I had to do to set up so that I could write up my own readme properly later on.
2. Solve 'path of closest length' problem
   (a) Have not started research yet

Week 7 (13th - 19th):

x Solve the 'path of shortest length' problem
   (a) Finish setting up GraphHopper so that I can get screenshots of the code!!

1. Solve 'path of closest length' problem
   (a) Start researching formulas that I can include in the report!!
   (b) Since it is NP (SubsetSum), need to find a way to simplify the problem into something I can efficiently solve

x Start draft 1 of the progress report
   (a) Do something about the FDD model
   (b) Finish finding data sets
   (c) Define measure of success
      i. based on completing tasks

Week 8 (20th - 26th):

x Finish Progress Report
   (a) Met with Nathan about the draft

3 Problems I have encountered

1. Komoot is closed source... needed to find an alternative data source for terrain data
2. Realised GraphHopper won't let you increase the priority on edges, only decrease the priority of other edges. this will impact my formulas later
3. Poor documentation makes it hard to use API
4. UI would only let you change road_environment and road_class

4 Promises I made in Report

1. Use an informed search algorithm, potentially over long distances (flexible mode)
   (a) use `distance_influence` to prioritise speed over distance (30?)
   (b) Research how to enter a custom model into the local instance

Moving forward

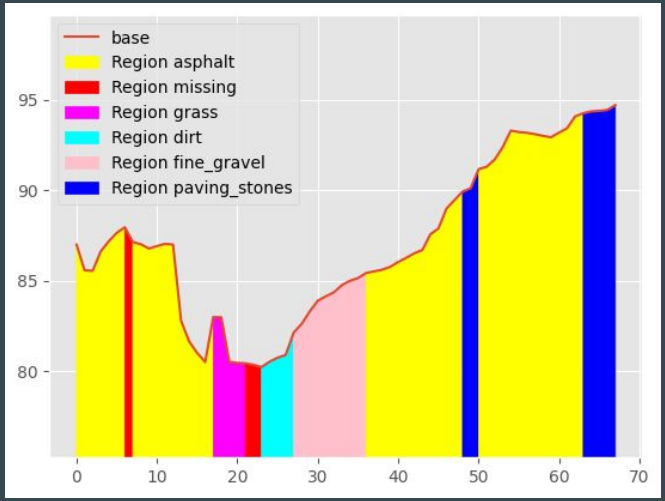
- Complete analysis for Slope, Surface, and Distance Influence Heuristics
- This will inform how the wheelchair vehicle is finalised

Future work

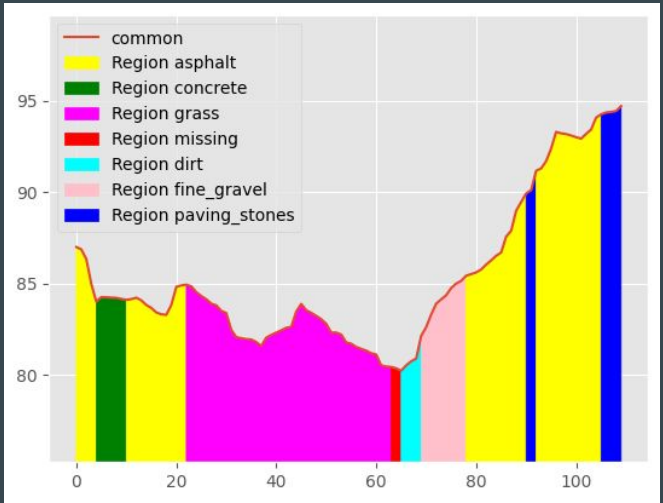
- Analyse how different heuristics affect each other (intersection)
- Incorporate functional metadata to analyse dynamic costs
- Perform analysis on multiple vehicle types
- Learn from user's data and what routes they prefer to find the heuristic values that best meet their needs

Questions?

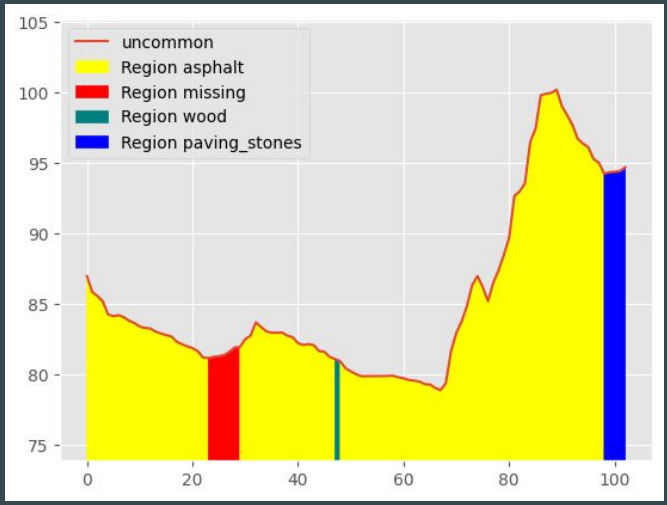
Surface against Elevation



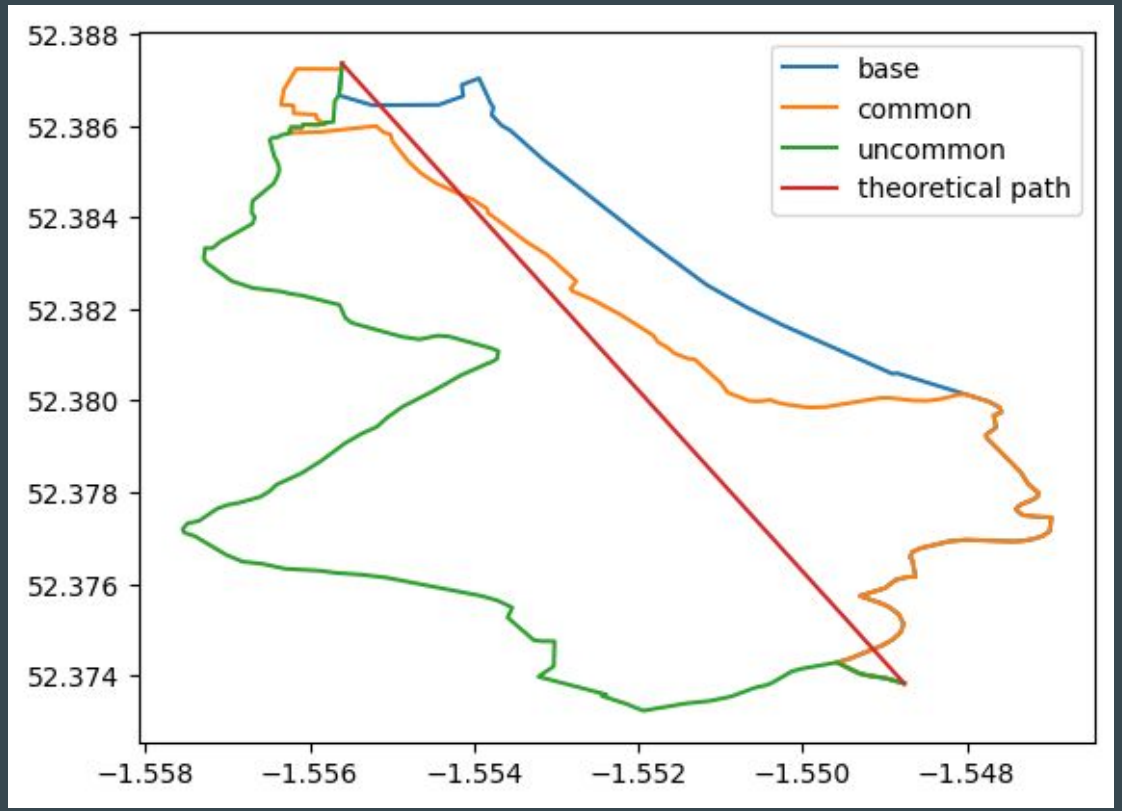
restrictions on ASPHALT

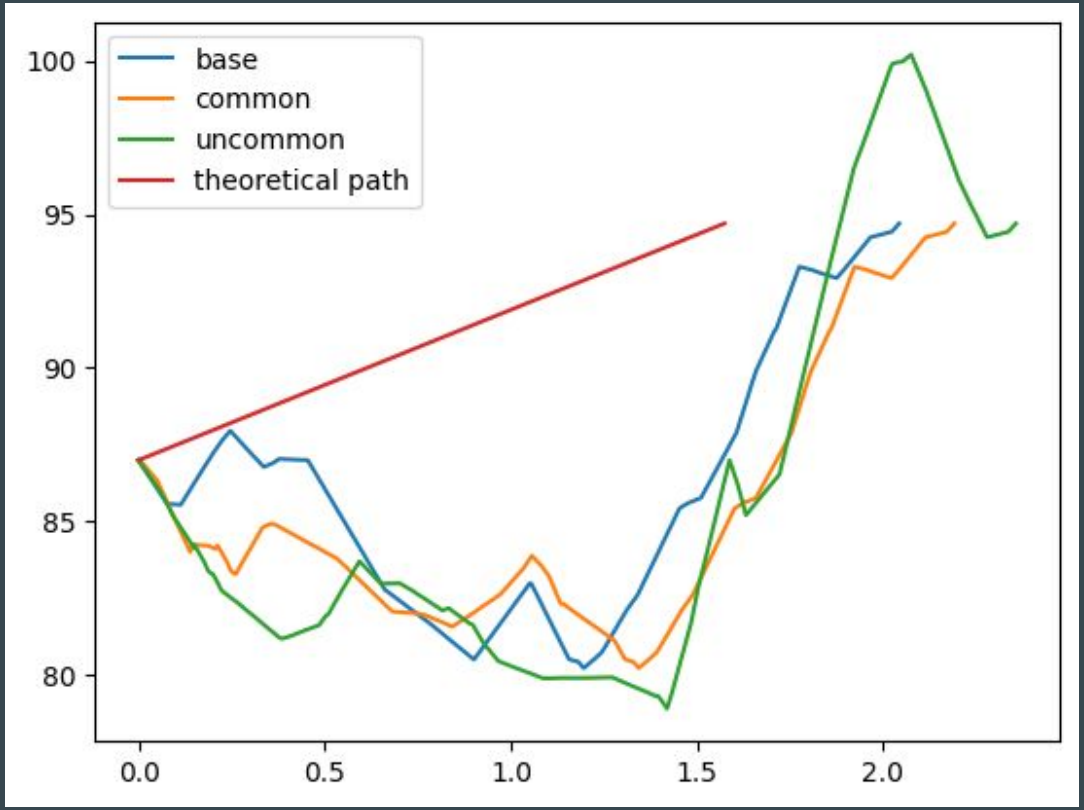


restrictions on DIRT

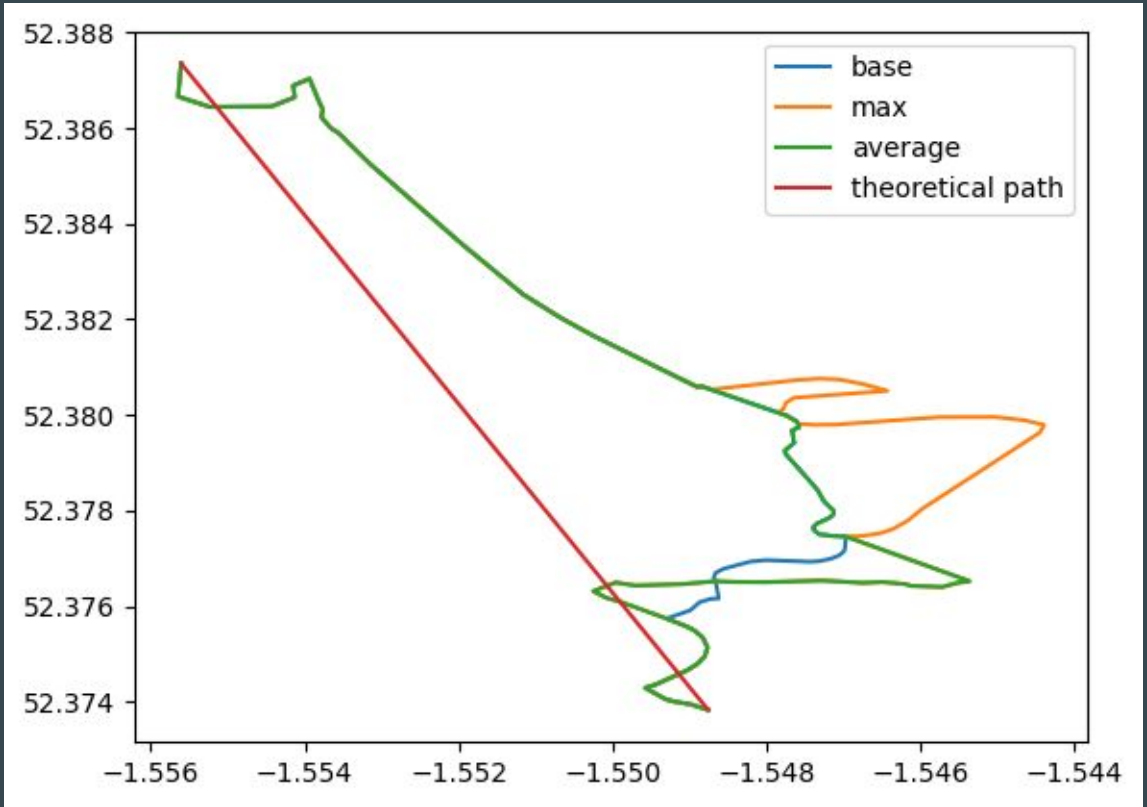


common vs uncommon

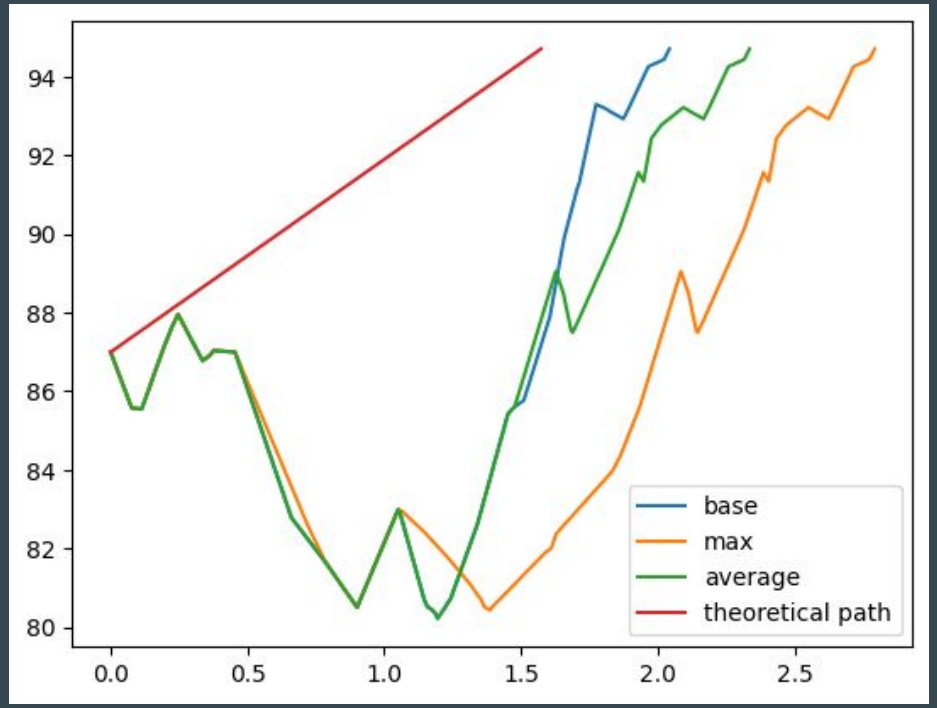


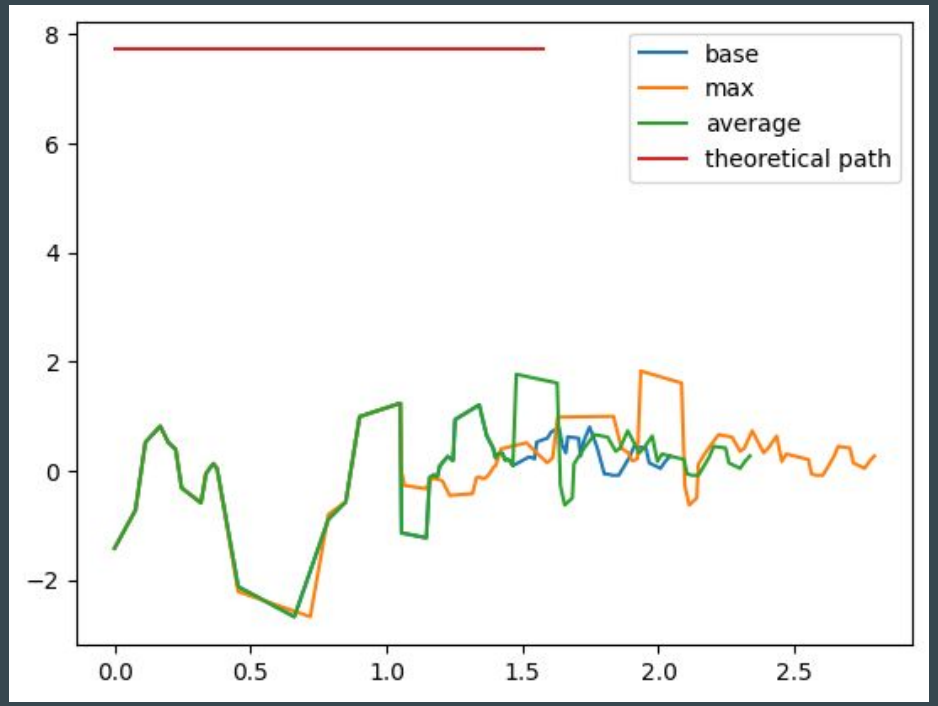

base
common
uncommon
theoretical path
100
95
90
85
80
0.0
0.5
1.0
1.5
2.0

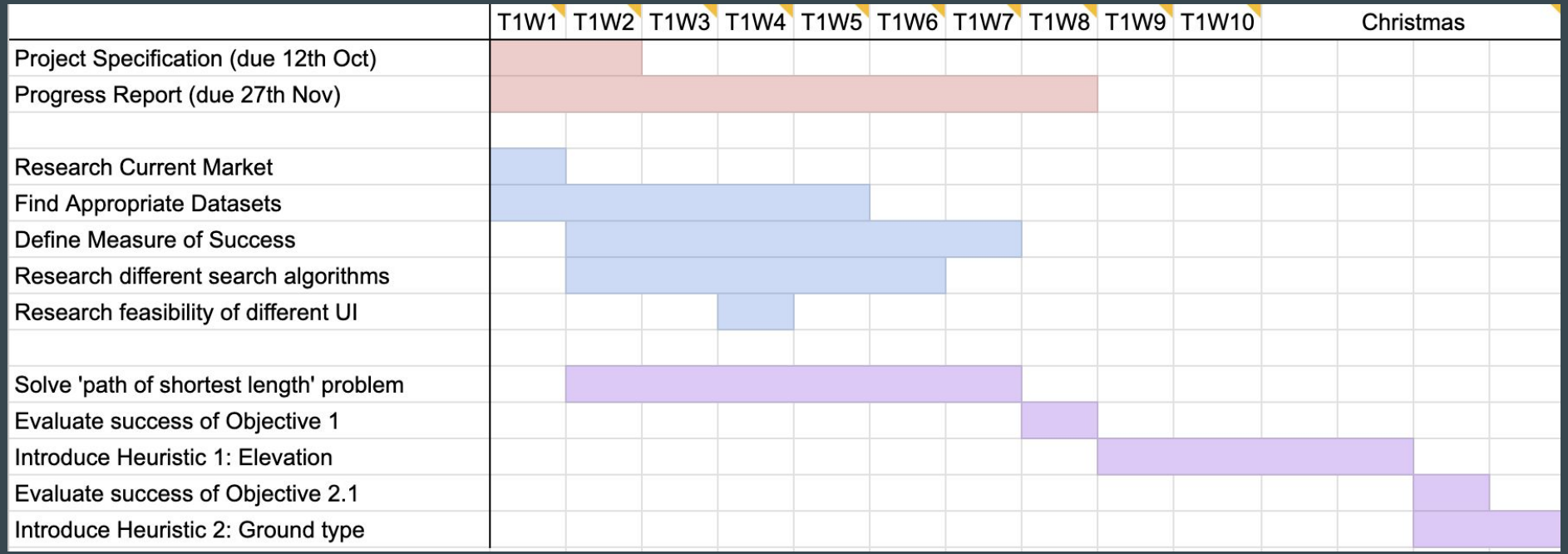
average_slope vs max_slope



average_slope vs max_slope






T1W1
T1W2
T1W3
T1W4
T1W5
T1W6
T1W7
T1W8
T1W9
T1W10
Christmas
Project Specification (due 12th Oct)
Progress Report (due 27th Nov)
Research Current Market
Find Appropriate Datasets
Define Measure of Success
Research different search algorithms
Research feasibility of different UI
Solve 'path of shortest length' problem
Evaluate success of Objective 1
Introduce Heuristic 1: Elevation
Evaluate success of Objective 2.1
Introduce Heuristic 2: Ground type

	T2W1	T2W2	T2W3	T2W4	T2W5	T2W6	T2W7	T2W8	T2W9	T2W10	Easter	T3W1	T3W2
Presentation (due 6th Mar 11:30)													
Final Report (due 30th Apr)													
Introduce Heuristic 2: Ground type													
Evaluate success of Objective 2.2													
Introduce Heuristic 3: Wheelchair accessibility													
Evaluate success of Objective 2.3													